



Project objectives and actors

Ingegneria del Software

2025/2026

Università di Trento

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Agenda



1.

2.

3.

Today

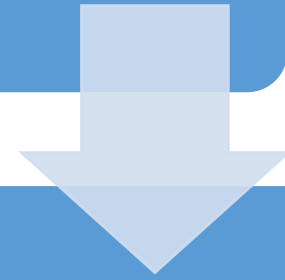
- Project objectives
- SWOT analysis
- System actors
- Structure of deliverable D1



Objective definition

From project idea to project objectives

Project idea



Project objectives

Objectives – app for fitness

O1: Improve the health and well-being of users	Provide a personalized experience to users, helping them to achieve their health and wellness goals, such as weight loss, improved fitness or stress reduction.
O2: Provide personalized training programs.	Provide personalized training programs, based on preferences and goals of users, so that they can obtain the best possible results.
O3: Monitor user progress	Monitor user progress, providing feedback and suggestions for improving performance and achieving goals.
O4: Provide nutrition information.	Provide information on nutrition and how to follow a balanced diet to support users' training programs.
O5: Provide information on mental health	The goal may be to provide information on mental health and stress, with suggestions on how to address these problems through physical activity and meditation.
O6: Provide social support	Provide social support to users, allowing them to connect with other sports enthusiasts and to share their experiences and successes.
O7: Promote outdoor physical activity	Promote outdoor physical activity, suggesting routes and activities to carry out outdoors, such as hiking, walking and cycling.
O8: Provide incentives and rewards	Incentivize users to achieve their goals through incentives and rewards, such as badges and virtual prizes.

Objectives – app for sustainable mobility

O1: Improve user experience	Create an intuitive and easy-to-use user experience, so that users can easily find the mobility options available, book trips and track their movements.
O2: Promote sustainable mobility	Encourage the use of sustainable means of transport, such as cycling, public transport and electric vehicles, providing information on available options and suggesting alternative routes to reduce environmental impact.
O3: Optimize route planning	Provide navigation aids in real time, suggesting the best routes for users based on traffic, weather and personal preferences.
O4: Increase road safety	Provide information on traffic conditions, roadworks and other potential obstacles along the route, as well as suggesting alternative routes in case of emergency.
O5: Create a community	Create a community of mobility users, who can share information on transport options available, suggest routes and give advice.
O6: Increase geographic coverage	Extend the app's reach into new geographic areas so users can use it wherever they are.
O7: Integrate with other app and services	Create integrations with other mobility services, such as car sharing, bike sharing and public transport, to provide a more complete and integrated mobility experience.
O8: Collect data and provide analysis.	Collect data on user mobility, to better understand their needs and habits, and use this data to improve the app and overall transportation planning.



SWOT analysis

SWOT analysis

- A technique for assessing **Strengths**, **Weaknesses**, **Opportunities** and **Threats** of a business
- Once specified the objective(s) of your project, it helps identify internal and external factors that are favorable and unfavorable to achieve that objective
- Internal aspects are evaluated by focusing on strengths and weaknesses
- External factors are evaluated by focusing on opportunities and threats

SWOT analysis



SWOT analysis – app for fitness

Strengths

- Customers interested in healthcare and wellness
- Increased awareness about healthy lifestyle
- Market increase of fitness app and devices

Weaknesses

- High costs of fitness programs and tools
- Limited access to fitness programs and tools for certain users
- Market opportunities focused on few geographic areas

Opportunities

- Global fitness market
- Integration of technology in fitness programs
- Increased awareness about the benefits of fitness at both physical and mental level

Threats

- Increased competitors on the fitness markets
- Reduction of the availability of public fundings for fitness and sporting activities
- Increase of the competitors at low cost

SWOT analysis – app for sustainable mobility

Strengths

- Growing interest in sustainable mobility and reduction of greenhouse gas emissions
- Presence of infrastructures for sustainable mobility such as cycle paths, charging stations for electric vehicles, etc.
- Advanced technologies such as sensors, artificial intelligence and the internet of things (IoT)

Weaknesses

- Low adoption of sustainable technologies and sustainable behaviors among some user segments
- Difficulties in collecting data on sustainable mobility
- Dependence on existing infrastructure, which may be inadequate or lacking in some areas

Opportunities

- Growth in demand for sustainable mobility services and innovative mobility solutions
- Possibility of collaboration with public bodies, companies and non-governmental organizations to promote sustainable mobility
- Use advanced data and analytics to improve user experience and optimize route planning

Threats

- Regulations and public policies that may limit the use and dissemination of the application
- Uncertainty about funding and sustainable business models for the application
- Possible lack of interoperability with other already consolidated sustainable mobility services



System actors

System actors

- The actors of a software system are the entities or components that interact with the system itself.
- Both users and other software systems, hardware or internal components of the system itself.

Steps for identifying system actors

1. Identify users

- Who will use the system?

2. Identify external systems

- Are there other software or hardware systems that will integrate with the system?

3. Define the roles of the actors

- It is important to define the roles that each actor has within the system.

4. Analyze the interactions of the actors

- For each actor, it is important to understand how he will interact with the system and which ones information he will need to view and enter into the system.

5. Identify permissions

- To ensure system security, it is important to define permissions for each actor, so that they can only access the information and features to which they are entitled.

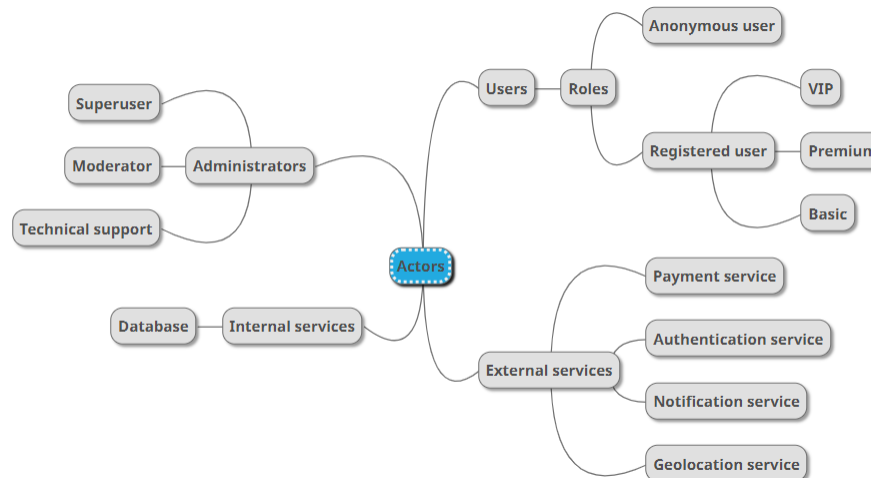
Mind maps



- An easy way to brainstorm thoughts without worrying about structure
- A diagram for representing tasks, words, concepts around a central concept or subject
- Based on the study of the brain's functionality, it is used to stimulate and challenge it
- A highly effective way of getting information in and out of your brain
- A creative and logical means of note-taking and note-making for mapping out ideas

Mind Map for identifying the system actors

- A mind map is a useful support for defining the actors the system and their roles.
- It also helps the team to understand system actions/functionalities for each role.
- For instance, <https://www.mindmup.com/>



Deliverable D1 (27-03-2026)

D1

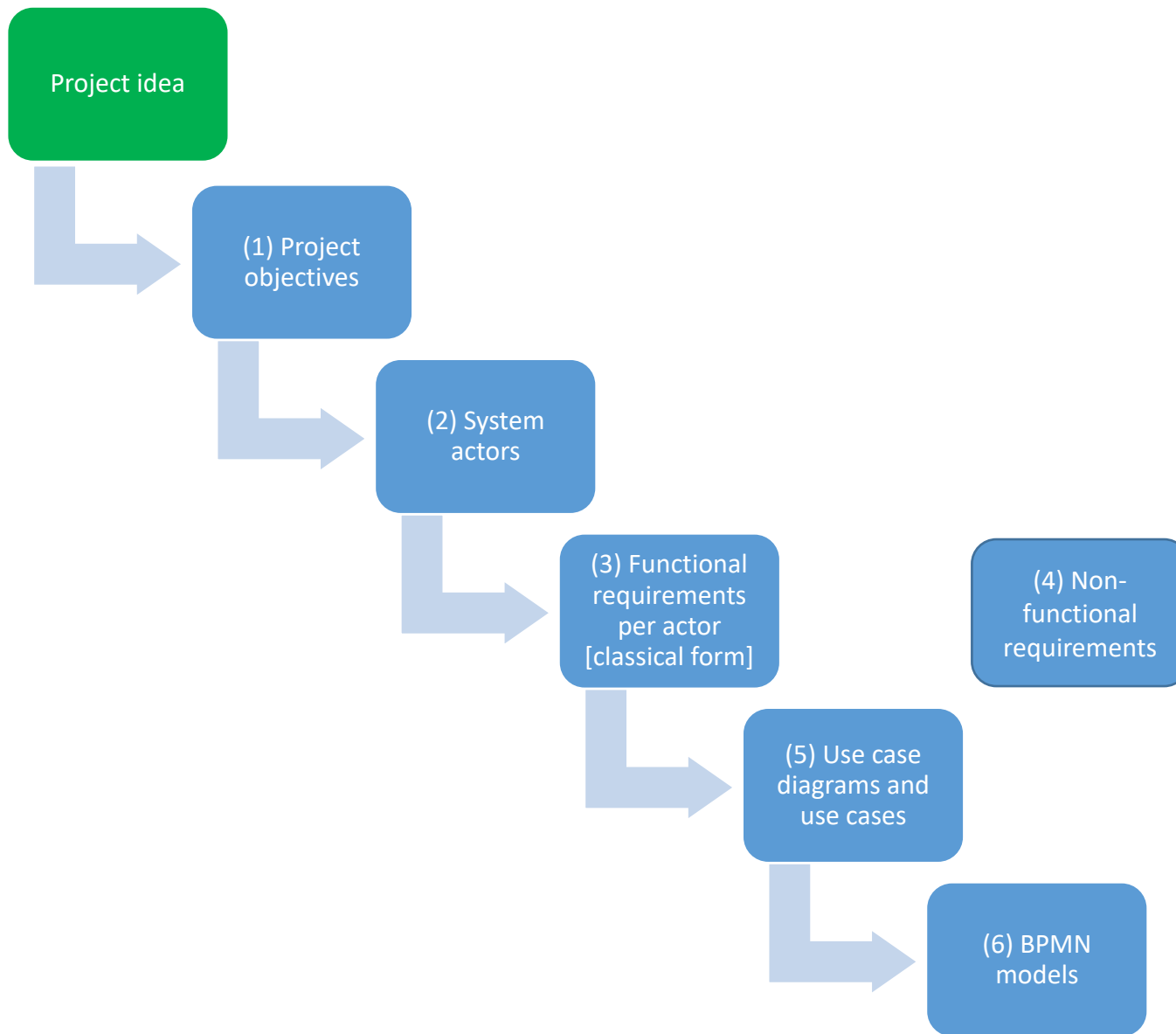
Project
objectives

System actors

Requirements
(functional
and non-
functional)

Use case and
BPMN
diagrams

Deliverable D1



Some tools

- Actor mindmap → Mindmup
(<https://app.mindmup.com/>)
- BPMN modelling → Camunda modeler
(<https://camunda.com/download/modeler/>)
- UML modelling → Eclipse papyrus
(<https://eclipse.dev/papyrus/download.html>)

Summary

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SUMMARY

